END SEMESTER EXAMINATION April'2025

DEPARTMENT OF MATHEMATICS & COMPUTING TECHNOLOGY

Subject: Engg. Maths II (Prob.& Stat) [**MA24102**_Backlog(MA24101)] *Time:3hrs*

Course : B.Tech (CSE I)+ backlog *F.M.: 60*

Semester: 2nd sem

Answer any Five questions

1(a)If a discrete random variable X take the value 0,1,2,.....n, with probabilities $\frac{c_0^n}{2^n}, \frac{c_1^n}{2^n}, \frac{c_2^n}{2^n}, \ldots\ldots\ldots, \frac{c_n^n}{2^n}$ respectively, show that the mean and variance of X is $\frac{n}{2}$ and $\frac{n}{4}$ respectively.

(b) The lifetime of a certain brand of an electric bulb may be consider a RV with mean 1200 hours and standard deviation 250 hours. Find the probability, using central limit theorem, that the average lifetime of 60 bulbs exceeds 1250 hours.

[*LOS be 5%* , $P(Z = 1.55) \approx 0.4394$]

2(a)Experience has shown that 20% of a manufactured product is of top quality. In one day's production of 400 articles, 50 are the top quality. Show that either the production of the day chosen was not a representative sample or the hypothesis of 20% was wrong. Based on the particular day's production, find also the 95% confidence limit for the percentage of top quality product. [*LOS be 5%* , $Z_\alpha = 1.96$]

(b) 15.5% of a random sample of 1600 undergraduates were smokers, whereas 20% of a random sample of 900 postgraduates were smokers in a state. Can we conclude that less number of undergraduates are smokers than the postgraduates?

[*LOS be 5%*, $Z_\alpha = -1.645$]

3(a)The mean weight obtained from a random sample of size 100 is 64 gms. The S.D. of the weight distribution of the population is 3 gms. Test the statement that the mean weight of the population is 67 gms at 5% level of significance. Also set up 99% confidence limit of the mean weight of the population. [*LOS be 5%*, $Z_\alpha = 1.96$]

(b)For sample I, $n_1 = 1000$, $\sum x = 49000$, $\sum(x - \bar{x}_1)^2 = 7,84,000$.
For sample II $n_2 = 1500, \sum x = 70500, \sum(x - \bar{x}_2)^2 = 2,400,000$.
Discuss the significance of the difference of the sample means. [*LOS be 5%*, $Z_\alpha = 1.96$]

4(a)A manufacturer of electric bulbs, according to a certain process, find the S.D, of life of lamps in a smaller variation in the life of lamps. In adopting new process, a sample of 150 bulbs gave an S.D. of 95 h. Is the manufacturer justified in changing the process?

[LOS be 5%, $Z_\alpha = 1.96$]

(b)The S.D. of a random sample of 1000 is found to be 2.6 and the S.D. of another random sample of 500 is 2.7. Assuming the sample to be independent, find whether the two sample could have come from populations with the same S.D. [LOS be 5%, $Z_\alpha = 1.96$]

[P.T.O]

5(a) A random sample of size 16 has 53 as mean. The sum of squares of the derivation from mean is 135. Can this sample be regarded as taken from the population having 55 as mean? Obtain 95% and 99% confidence limits of the mean of the population.

$[t_{0.05} = 1.753 \text{ for } d.f\ \vartheta = 15;\ t_{0.01} = 2.602 \text{ for } d.f\ \vartheta = 15]$

(b) Below are given the gain in weight (in lbs) of pigs fed on two diets A and B.

	Gain in weight
Diets A	25, 32, 30, 34, 24, 14, 32, 24, 30, 31, 35, 25
Diets B	44, 34, 22, 10, 47, 31, 40, 30, 32, 35, 18, 21, 35, 29, 22

Test, if the two diets differ significantly as regard their effect on increases in weight.

$[t_{0.05} = 2.06, \text{for } d.f\ \vartheta = 25]$

6(a) Two random samples drawn from two normal populations have the variable values as below:

Sample I	19, 17, 16, 28, 22, 23, 19, 24, 26
Sample II	28, 32, 40, 37, 30, 35, 40, 28, 41, 45, 30, 36

Obtain the estimate of variance of the population and test whether the two population have the same variance.

$[F = 3.315 \text{ for } d.f\ \vartheta_2 = 11, \vartheta_1 = 8\ at\ 5\%\ LOS]$

(b) Four methods are under development for making discs of super conducting material. 50 discs are made by each method and they are checked for super conducting when cooled with liquid nitrogen. The table as follows:

	Method I	Method 2	Method 3	Method 4
Super conductors	31	42	22	25
Failures	19	8	28	25
Total	50	50	50	50

Perform Chi-square test with $\propto = 0.05$. If there is a significant difference between the proportions of super conductor produced, plot the individual confidence intervals.

$[\chi^2 = 7.815 \text{ for } d.f\ \vartheta = 3\ at\ 5\%\ LOS]$

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